

Question number	Answer	Additional guidance	Mark
2(a)(i)	choose an item: D (regulate the transcription of genes) is the correct answer A is not the correct answer as gibberellins do not hydrolyse starch to glucose B is not the correct answer as gibberellins do not inhibit elongation of cells C is not the correct answer as gibberellins do not regulate the translation of DNA		(1)

Question number	Answer	Additional guidance	Mark
2(a)(ii)	choose an item: B (adrenaline) is the correct answer A is not the correct answer as adenine is not a hormone C is not the correct answer as amylopectin is not a hormone D is not the correct answer as antihypertensive is not a hormone		(1)

Question number	Answer	Additional guidance	Mark
2(b)(i)	An answer that includes the following points: • as TBT concentration increases testosterone concentration (in blood) decreases / converse (1) • as TBT concentration increases up to 50 mg kg⁻¹ oestradiol concentration (in blood) increases and then decreases {above 50 mg kg⁻¹ / at 100 mg kg⁻¹} (1) • concentration of testosterone is higher than concentration of oestradiol (with no TBT / in all concentrations) (1)	ignore concentration of corn oil with no ref to TBT ACCEPT oestrogen for oestradiol ACCEPT negative correlation between TBT and testosterone concentrations ACCEPT TBT decreases testosterone (concentration) ACCEPT as TBT concentration increases, oestradiol concentration increases and then decreases ACCEPT as TBT concentration increases up to 50 mg kg⁻¹ oestradiol concentration (in blood) increases / positive correlation up to 50 mg kg⁻¹	(3)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	A description that gives three of the following points - steroid hormone diffuses {across the cell membrane / into cell / into nucleus} (1) (steroid hormones) bind to {receptors / proteins} (inside cell / nucleus) (1) - which {act as a / become / interact with} transcription factors (1) - causing {activation / repression} of gene (transcription) (1)	ACCEPT steroids for steroid hormone ACCEPT steroid hormone enters {cell / nucleus} ACCEPT pass through for diffuse reject binding to cell membrane receptors ACCEPT forms hormone-receptor complex ACCEPT bind to {promoter region / RNA polymerase} ignore binds to DNA ACCEPT {enable / inhibit} RNA polymerase binding ACCEPT switching genes {on / off} ACCEPT {allowing / preventing} {transcription / formation of mRNA}	(3)

